

Teaching Introductory Statistics and Assessing Learning

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bit.ly/PTT_26_TG

[Video](#)



Five years from now, what do you want your intro stats students to remember?

Take 1 minute to think on it on your own

Then, turn to a neighbor and discuss each others answers (3 min).

Same or different answers? Surprised?

Think pair share



Context

Cornell College

- Small private liberal arts college, 1000 students
- Iowa
- Majors in statistics and data science
- Department of Mathematics and Statistics, 4 members
- 30ish majors in department, more minors
- Not highly selective



Context

One course at a time

- Students take and I teach, one class at a time
- 18, four hour class days, 2 hours morning/afternoon
- 6 courses a year, 3cr equivalent but 4cr teaching time
- No course support beyond librarians



Experience

My teaching experience (specifically intro stats)

- 5 years teaching 3 courses a year as a graduate student
- 6 years at Cornell College including 13 sections of introductory statistics
- In person instruction



My Introductory Statistics Course

- Book: Statistics: Unlocking the Power of Data, 3rd Edition
- Students: a mix of STEM students and other students taking the course for their quantitative reasoning requirement
- Software: StatKey (free online, Lock5 Book), Minitab/JMP
 - *No, I don't use R in this course*
- More details in Syllabus included in shared folder



Examples



Will You Teach the Way You Were Taught?

- Many of us learned intro stats as a sequence of procedures in lectures
 - I learned from a ‘robot’
- Active learning is ubiquitous and educational research supports its continued use
 - Freeman et al., 2014
 - Theobald et al, 2020



Guidelines for Assessment and Instruction in Statistics Education (GAISE)

- I am not the expert
- The people who wrote the GAISE report are
- Ten recommendations, revised 2025
- Real data with context and purpose. Ethical conduct. Inclusive practice. Concepts over algebra. Software. Active learning, and more.
- Tons of activities and other materials
- <https://amstat.quarto.pub/college-gaise/>



Activity 1: Day one, “marbles”

- First moment of class I drop a loud bag of marbles on the front table
- Students draw marbles and try to estimate an unknown proportion of green marbles
- They do this multiple times with different sample sizes
- I ask them to reason, from intuition about sample size impacts on estimates, variability between point estimates, and the lack of reliability of a point estimate



Example: The what matters most to you survey

- I use a [Google Form](#) to see what is “important” to the class
- Students are often shocked that a majority of students put academics first
- This survey feeds later discussions
- *GAISE 4: real data with context and purpose, chosen to be meaningful to these students.*



Activity 3: LEGO

- Anna Peterson and Laura Ziegler (2021) published a paper, activity, and data about LEGOs (JSDSE)
- Data includes some clean and some messy columns
- My modified activity works well for my introductory statistics classroom
- Reviews content on descriptive statistics, and, helps me show students what an EDA entails (often many plots!)



Activity 4: Sampling distributions, in M&Ms

- Sampling distributions are hard for students!
- Every student takes a sample of M&Ms and computes a proportion
- The class creates a dot plot on the board, their first sampling distribution
- (I think) that students connect each dot on the board with the physical “bag” of M&Ms they received



Activity 5: Bootstrapping, in M&Ms

- Lock5 teaching simulation based inference first - Bootstrapping
- Each group is given only one sample (no class effort)
- They are tasked with finding a way to create a sampling distribution using one sample
- I let them struggle, for a while
- Eventually I give them the fine details, they practice it by hand a few times, and then use software
- Students remember this activity well



Discussing Ethics, Race, and Gender in Intro Stats Class?

- Common statistics examples discuss only 2 genders and/or sexes
- Discussions about gender come up organically in my introductory statistics class often
- We have a class discussion early on, and, reference it often
- I like to mention the JEDI Outreach group, and an [article by Dr. Suzanne Thornton](#) (2022)
- I've done a better job incorporating these into my intro to DS course
- *GAISE 7: responsible and ethical conduct in the collection, use, and analysis of data.*



Assessment



No single instrument tells you everything

- GAISE 9: use a variety of formative and summative assessments to improve teaching and learning.
- Formative assessments are regular, and inform your teaching and which students are keeping up, can be daily, or, even in the middle of class
- Summative assessments are traditionally tests/exams, including the final exam
- Quizzes, projects, and exams all assess different skills, knowledge, competencies



Quizzes

- About 7 quizzes across an 18-day course
- Frequent and low stakes, so no single one can sink a student
- Gives me a rough idea who is behind, and if the class was challenged by a particular topic



Projects

- GAISE 2: communicate results in writing and in speech, with the scope and limitations of the conclusion.
- Students choose between 3 different datasets, do a short literature review, write their own research questions, perform the statistical analysis, and give a conclusion



Project Rubric

- [Project 2 Rubric Example](#): criteria, levels, point values, exactly as students see it.
- Three things to point at when you show it: how few criteria there are, that students get it before they start, and that communication is scored alongside correctness.
- Derived from work of Dr. Alana Unfried



Bad Test Question Example

Rosiglitazone is the active ingredient in the controversial type 2 diabetes medicine Avandia and has been linked to increased risk of serious cardiovascular problems such as stroke, heart failure, and death. A common alternative treatment is Pioglitazone, the active ingredient in Actos. In a nationwide retrospective observational study of 227,571 Medicare beneficiaries aged 65 years or older, 2,593 of the 67,593 patients using Rosiglitazone and 5,386 of the 159,978 using Pioglitazone had serious cardiovascular problems.



Let's Pick This Apart

To yourself: What is this question actually testing? What do you think I wanted to test?

Once you have an answer, talk to your neighbor again! How could it be improved?



Better Test Question Example

In December 2025, the FDA approved a new drug to treat Vomiting Induced Motion. They compared the drug to a placebo in a clinical trial. The results are summarized below.



More on Assessment

- Consider what the students need for learning AND what's practical.
- Rubrics help with consistency
- Alternative grading is popular and there are resources and communities to help
- Feedback students do not read costs the same hours as feedback they do. *I'm going to be thinking about this this fall.*



General Thoughts



What did you not really understand until graduate school?

- TALK TO YOUR NEIGHBOR - 2 minutes
- A concept from your own intro or undergraduate statistics classes.
- Then: what would have helped you get it sooner?



What Has Surprised Me

- Engagement is work. I spend a lot of energy convincing students they can do statistics, and, making them love it
- Even the most disengaged student pays attention if the data interests them
- Students arrive more aware than they used to be that they need statistics



Advice

- Teach the same prep twice before you redesign it. The first run is data collection.
- Do not build from scratch.
- One deliberate change per term. Small incremental change is progress, and it is safer for student learning
- Low tech is OK
- Open source when possible



Use Existing Resources

- Lock5 authors wrote slides, videos, and activities to go with the book. They ship in WileyPLUS.
- Most of the included activities today come from Lock5
- Free: CAUSEweb, OpenIntro, the GAISE reports, Intro to Modern Statistics
- Ask your department for materials to start with.
- Adapting someone else's material, with credit, is not cutting corners. It is the job done well.
- [Section of Statistics and Data Science Education links](#)



Questions & Discussion

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